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Semiconductor package

Abstract

A semiconductor package is provided. The semiconductor package includes: a first redistribution substrate; a semiconductor chip provided on the first redistribution substrate; a molding layer provided on the first redistribution substrate and the semiconductor chip; and a second redistribution substrate provided on the molding layer. The second redistribution substrate includes: redistribution patterns spaced apart from one another; a first dummy conductive pattern spaced apart from the redistribution patterns; an insulating layer provided on the first dummy conductive pattern; and a marking metal layer provided on the insulating layer and spaced apart from the first dummy conductive pattern. Sidewalls of the marking metal layer overlap the first dummy conductive pattern along a vertical direction perpendicular to an upper surface of the first redistribution substrate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0039150, filed on Mar. 29, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.
BACKGROUND
(2) The present disclosure relates to a semiconductor package, and more particularly, relates to a semiconductor package including a redistribution substrate and a method of manufacturing the same.
(3) A semiconductor package may be implemented in a form suitable for use in an electronic product using an integrated circuit chip. In a semiconductor package, a semiconductor chip may be mounted on a printed circuit board and electrically connected thereto using bonding wires or bumps. With development of electronics industry, various studies are being conducted for reliability improvement, high integration, and miniaturization of the semiconductor package.
SUMMARY
(4) One or more example embodiments provide a semiconductor package having improved reliability and durability.
(5) According to an example embodiment, a semiconductor package includes: a first redistribution substrate; a semiconductor chip provided on the first redistribution substrate; a molding layer provided on the first redistribution substrate and the semiconductor chip; and a second redistribution substrate provided on the molding layer. The second redistribution substrate includes: redistribution patterns spaced apart from one another; a first dummy conductive pattern spaced apart from the redistribution patterns; an insulating layer provided on the first dummy conductive pattern; and a marking metal layer provided on the insulating layer and spaced apart from the first dummy conductive pattern. Sidewalls of the marking metal layer overlap the first dummy conductive pattern along a vertical direction perpendicular to an upper surface of the first redistribution substrate.
(6) According to an example embodiment, a semiconductor package includes: a first redistribution substrate; a semiconductor chip provided on the first redistribution substrate; a molding layer provided on the first redistribution substrate and the semiconductor chip; and a second redistribution substrate provided on the molding layer. The second redistribution substrate includes: a redistribution pattern; a first dummy conductive pattern insulated from the redistribution pattern; a second dummy conductive pattern insulated from the redistribution pattern; a third dummy conductive pattern insulated from the redistribution pattern; and a marking metal layer provided on the second dummy conductive pattern. The first dummy conductive pattern is provided between the second dummy conductive pattern and the third dummy conductive pattern. The marking metal layer overlaps a first portion of the first dummy conductive pattern along a vertical direction perpendicular to an upper surface of the first redistribution substrate, and is offset from a second portion of the first dummy conductive pattern along the vertical direction.
(7) According to an example embodiment, a semiconductor package includes: a first redistribution substrate including a first insulating layer, a first seed pattern, and a first redistribution pattern; a solder ball provided on a bottom surface of the first redistribution substrate; a semiconductor chip provided on a top surface of the first redistribution substrate; conductive structures provided on the top surface of the first redistribution substrate and spaced apart from the semiconductor chip along a horizontal direction parallel to an upper surface of the first redistribution substrate; a molding layer provided between the semiconductor chip and the conductive structures, and on the semiconductor chip; and a second redistribution substrate provided on the molding layer. The second redistribution substrate includes: second redistribution patterns electrically connected to the conductive structures; second redistribution pads provided on and electrically connected to the second redistribution patterns; a dummy conductive pattern spaced apart from the second

redistribution patterns along the horizontal direction; and a marking metal layer spaced apart from the second redistribution pads along the horizontal direction. The dummy conductive pattern includes a first dummy conductive pattern, a second dummy conductive pattern, and a third dummy conductive pattern that are spaced apart from one another along the horizontal direction. The marking metal layer overlaps a first portion of the first dummy conductive pattern and the second dummy conductive pattern along a vertical direction perpendicular to the upper surface of the first redistribution substrate. The marking metal layer is offset from a second portion of the first dummy conductive pattern and the third dummy conductive pattern along the vertical direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects and features will be more clearly understood from the following description of example embodiments taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1**A is a plan view illustrating a second redistribution substrate of a semiconductor package according to example embodiments;

(3) FIG. **1**B is a plan view illustrating an arrangement of dummy conductive patterns and second redistribution patterns of a second redistribution substrate according to example embodiments;

(4) FIG. **1**C is a plan view illustrating arrangement of a marking metal layer and second redistribution pads of a second redistribution substrate according to example embodiments;

(5) FIG. **1**D is a cross-sectional view taken along line I-II of FIG. **1**A;

(6) FIG. **1**E is an enlarged view of region "III" of FIG. **1**D;

(7) FIG. **1**F is a view for illustrating a related marking metal layer;

(8) FIG. **1**G is a view for illustrating a marking metal layer and a second redistribution pad according to example embodiments;

(9) FIG. **2**A is a plan view illustrating a marking metal layer according to example embodiments;

(10) FIG. **2**B is a cross-section taken along line IV-V line of FIG. **2**A;

(11) FIG. **2**C is a view for illustrating a marking metal layer according to example embodiments;

(12) FIG. **2**D is a view for illustrating a marking metal layer according to example embodiments;

(13) FIG. **2**E is a view for illustrating a marking metal layer according to example embodiments;

(14) FIG. **3**A is a diagram for illustrating a semiconductor package according to example embodiments;

(15) FIG. **3**B is a diagram for illustrating a semiconductor package according to example embodiments;

(16) FIG. **3**C is a diagram for illustrating a semiconductor package according to example embodiments;

(17) FIG. **4** is a view for illustrating a semiconductor package according to example embodiments; and

(18) FIGS. **5**A, **5**B, **5**C, **5**D, **5**E, **5**F, **5**G, **5**H, SI, **5**J and **5**K are diagrams for illustrating a method of manufacturing a semiconductor package according to example embodiments.

DETAILED DESCRIPTION

(19) Example embodiments will be described more fully hereinafter with reference to the accompanying drawings. The same reference numerals may refer to the same elements throughout. Expressions such as "at least one of," when preceding a list of elements, modify the entire list of elements and do not modify the individual elements of the list. For example, the expression, "at least one of a, b, and c," should be understood as including only a, only b, only c, both a and b, both a and c, both b and c, or all of a, b, and c. It will be understood that when an element or layer is referred to as being "on," "connected to" or "coupled to" another element or layer, it can be

directly on, connected or coupled to the other element or layer, or intervening elements or layers may be present. By contrast, when an element is referred to as being "directly on," "directly connected to" or "directly coupled to" another element or layer, there are no intervening elements or layers present.

(20) FIG. **1**A is a plan view illustrating a second redistribution substrate of a semiconductor package according to example embodiments. FIG. **1**B is a plan view illustrating an arrangement of dummy conductive patterns and second redistribution patterns of a second redistribution substrate according to example embodiments. FIG. **1**C is a plan view illustrating arrangement of a marking metal layer and second redistribution pads of a second redistribution substrate according to example embodiments FIG. **1**D is a cross-sectional view taken along line I-II of FIG. **1**A. FIG. **1**E is an enlarged view of region "III" of FIG. **1**D. FIG. **1**F is a view for illustrating a related marking metal layer. FIG. **1**D corresponds to a cross-section taken along line I-II of FIG. **1**B and a cross-section taken along line I-II of FIG. **1**C.

(21) Referring to FIGS. **1**A to **1**E, a semiconductor package **10** may be a lower package. The semiconductor package **10** may include a first redistribution substrate **100**, solder balls **500**, a semiconductor chip **200**, conductive structures **350**, a molding layer **400**, and a second redistribution substrate **600**.

(22) As shown in FIG. **1**D, the first redistribution substrate **100** may include a first insulating layer **101**, first redistribution patterns **130**, first seed patterns **135**, and first redistribution pads **150**. The first insulating layer **101** may include, for example, an organic material such as a photo-imageable dielectric (PID) material. The photosensitive insulating material may include, for example, at least one of a photosensitive polyimide, polybenzoxazole, a phenol-based polymer, and a benzocyclobutene-based polymer. The first redistribution substrate **100** may include a plurality of stacked first insulating layers **101**. The number of the first insulating layers **101** that are stacked may be variously modified. For example, the plurality of first insulating layers **101** may include the same material. Interfaces between adjacent first insulating layers **101** may not be distinguished. For example, the plurality of first insulating layers **101** may include different materials. Interfaces between adjacent first insulating layers **101** may be distinguished.

(23) The first redistribution substrate **100** may further include under bump patterns **120**. The under bump patterns **120** may be provided in the lowermost first insulating layer **101**. Bottom surfaces of the under bump patterns **120** may not be covered by the lowermost first insulating layer **101**. The under bump patterns **120** may function as pads of the solder balls **500**. The under bump patterns **120** may be laterally spaced apart from one another and may be electrically insulated from one another. When two components are laterally spaced apart, it may indicate that they are horizontally spaced apart. "Horizontal" may indicate being parallel to a bottom surface of the first redistribution substrate **100**. The bottom surface of the first redistribution substrate **100** may include a bottom surface of the lowermost first insulating layer **101** and the bottom surfaces of the under bump patterns **120**. The under bump patterns **120** may include a metal material such as copper.

(24) The first redistribution patterns **130** may be provided on the under bump patterns **120** and may be electrically connected to the under bump patterns **120**. The first redistribution patterns **130** may be laterally spaced apart from one another and may be electrically separated from one another. The first redistribution patterns **130** may be provided in the first insulating layers **101**. The first redistribution patterns **130** may include a metal such as copper. An electrical connection to the first redistribution substrate **100** may include an electrical connection to one of the first redistribution patterns **130**. The electrical connection of the two components to each other may include a direct connection or an indirect connection through another component.

(25) Each of the first redistribution patterns **130** may include a first via part and a first wiring part. In the present disclosure, a via part of a component may be a part for vertical connection, and a wiring part may be a part for horizontal connection. "Vertical" may indicate being perpendicular to the bottom surface of the first redistribution substrate **100**. The first via part may be provided in the

corresponding first insulating layer **101**. The first wiring part may be provided on the first via part and may be connected to the first via part without an interface.
(26) The first redistribution patterns **130** may include first lower redistribution patterns and first upper redistribution patterns. The first upper redistribution patterns may be disposed on the first lower redistribution patterns and may be connected to the first lower redistribution patterns, respectively. The number of the first redistribution patterns **130** stacked between the under bump patterns **120** and the first redistribution pads **150** is not limited to the drawings shown and may be variously modified.
(27) The first seed patterns **135** may be respectively disposed on bottom surfaces of the first redistribution patterns **130**. For example, each of the first seed patterns **135** may cover the bottom surface and the sidewall of the first via part of the corresponding first redistribution pattern **130** and a bottom surface of the first wiring part. Each of the first seed patterns **135** may not extend on the sidewall of the first wiring part of the corresponding first redistribution pattern **130**. The first seed patterns **135** may include a material different from that of the under bump patterns **120** and the first redistribution patterns **130**. For example, the first seed patterns **135** may include a conductive seed material. The conductive seed material may include copper, titanium, and/or alloys thereof. The first seed patterns **135** may function as barrier layers to prevent diffusion of a material included in the first redistribution patterns **130**.
(28) The first redistribution pads **150** may be disposed on the first redistribution patterns **130** to connect to the first redistribution patterns **130**. The first redistribution pads **150** may be laterally spaced apart from one another. Each of the first redistribution pads **150** may be connected to a corresponding under bump pattern **120** through the first upper redistribution pattern and the first lower redistribution pattern. The first redistribution patterns **130** may be provided, and thus the at least one first redistribution pad **150** may not be vertically aligned with the under bump pattern **120** electrically connected thereto. Accordingly, the arrangement of the first redistribution pads **150** may be designed more freely.
(29) The first redistribution pads **150** may be provided in the uppermost first insulating layer **101** and extend onto a top surface of the uppermost first insulating layer **101**. A lower part of each of the first redistribution pads **150** may be disposed in the uppermost first insulating layer **101**. An upper part of each of the first redistribution pads **150** may be provided on the lower part and may be connected to the lower part without an interface. The upper part of each of the first redistribution pads **150** may extend to the top surface of the uppermost first insulating layer **101**.
(30) First seed pads **155** may be respectively provided on bottom surfaces of the first redistribution pads **150**. As shown in FIG. **1**A, the first seed pads **155** may be respectively provided between the upper redistribution patterns of the first redistribution patterns **130** and the first redistribution pads **150**, and may extend between the uppermost first insulating layer **101** and the first redistribution pads **150**. The first seed pads **155** may include a material different from that of the first redistribution pads **150**. The first seed pads **155** may include, for example, a conductive seed material.
(31) The solder balls **500** may be disposed on the bottom surface of the first redistribution substrate **100**. For example, the solder balls **500** may be respectively disposed on the bottom surfaces of the under bump patterns **120** to be respectively connected to the under bump patterns **120**. The solder balls **500** may be electrically connected to the first redistribution patterns **130** through the under bump patterns **120**. The solder balls **500** may be electrically separated from one another. The solder balls **500** may include a solder material. The solder material may include, for example, tin, bismuth, lead, silver, or alloys thereof. The solder balls **500** may include a signal solder ball, a ground solder ball, and a power solder ball.
(32) The semiconductor chip **200** may be mounted on a top surface of the first redistribution substrate **100**. The semiconductor chip **200** may be disposed on the center region of the first redistribution substrate **100** in a plan view. The semiconductor chip **200** may be one of a logic chip,

a buffer chip, and a memory chip. For example, the semiconductor chip **200** may be a logic chip. The semiconductor chip **200** may include an ASIC chip or an application processor (AP) chip. The ASIC chip may include an application specific integrated circuit (ASIC). As another example, the semiconductor chip **200** may include a central processing unit (CPU) or a graphics processing unit (GPU).

(33) The semiconductor chip **200** may have top and bottom surfaces opposite to each other. The bottom surface of the semiconductor chip **200** may face the first redistribution substrate **100** and may be an active surface. The top surface of the semiconductor chip **200** may be an inactive surface. For example, the semiconductor chip **200** may include a semiconductor substrate, integrated circuits, a wiring layer, and chip pads **230**. The semiconductor substrate may include silicon, germanium, and/or silicon-germanium. The semiconductor substrate may be a silicon wafer. The integrated circuits may be adjacent to each other in the semiconductor chip **200**. The chip pads **230** may be provided on the bottom surface of the semiconductor chip **200**. The wiring layer may be provided between the integrated circuits and the chip pads **230**. The chip pads **230** may be connected to the integrated circuits through the wiring layer. When a component is electrically connected to the semiconductor chip **200**, it may indicate that it is electrically connected to the integrated circuits of the semiconductor chip **200** through the chip pads **230** of the semiconductor chip **200**. The chip pads **230** may include a metal such as aluminum, copper, and/or a combination thereof.

(34) As another example, the semiconductor chip **200** may include a plurality of lower chips. The lower chips may be horizontally spaced apart from one another. Alternatively, the lower chips may be vertically stacked on the first redistribution substrate **100**. Hereinafter, a single semiconductor chip **200** is illustrated and described for convenience, but example embodiments are not limited thereto.

(35) The semiconductor package **10** may further include conductive bumps **250**. The conductive bumps **250** may be interposed between the first redistribution substrate **100** and the semiconductor chip **200**. For example, the conductive bumps **250** may be provided between the corresponding first redistribution pads **150** and the chip pads **230** and may be connected to the first redistribution pads **150** and the chip pads **230**. Accordingly, the semiconductor chip **200** may be connected to the first redistribution substrate **100** through the conductive bumps **250**. The conductive bumps **250** may include solder balls. The conductive bumps **250** may include a solder material. The conductive bumps **250** may further include pillar patterns, and the pillar patterns may include a metal such as copper.

(36) The semiconductor package **10** may further include an underfill layer **410**. The underfill layer **410** may be provided in a gap region between the first redistribution substrate **100** and the semiconductor chip **200**, and may cover sidewalls of the conductive bumps **250**. The underfill layer **410** may include an insulating polymer such as an epoxy polymer.

(37) The conductive structures **350** may be disposed on the top surface of the first redistribution substrate **100** to connect to a corresponding one of the redistribution pads **150**. Accordingly, the conductive structures **350** may be connected to the first redistribution substrate **100**. The conductive structures **350** may be electrically connected to the solder balls **500** or the semiconductor chip **200** through the first redistribution substrate **100**. Metal pillars may be provided on the first redistribution substrate **100** to form conductive structures **350**. That is, the conductive structures **350** may be metal pillars.

(38) The conductive structures **350** may be laterally spaced apart from the semiconductor chip **200**. The conductive structures **350** may be laterally spaced apart from each other. The conductive structures **350** may be disposed on an edge region of the first redistribution substrate **100** in a plan view. The edge region of the first redistribution substrate **100** may be provided between the center region and sidewalls of the first redistribution substrate **100** in a plan view. The edge region of the first redistribution substrate **100** may surround the center region in a plan view.

(39) The conductive structures **350** may include signal carrying conductive structures and voltage supply conductive structures. The voltage may be a power supply voltage or a ground voltage.
(40) A molding layer **400** may be provided on the top surface of the first redistribution substrate **100** and may cover the semiconductor chip **200**. The molding layer **400** may further cover sidewalls of the conductive structures **350**. The molding layer **400** may be interposed between the semiconductor chip **200** and the conductive structures **350**. The molding layer **400** may include an insulating polymer such as an epoxy-based polymer.
(41) The second redistribution substrate **600** may be disposed on the semiconductor chip **200**, the molding layer **400**, and the conductive structures **350**. The second redistribution substrate **600** may include a second insulating layer **601**, second redistribution patterns **630**, second seed patterns **635**, a first dummy conductive pattern **641**, a second dummy conductive pattern **642**, a third dummy conductive pattern **643**, a second redistribution pad **650**, and a marking metal layer **660**.
(42) The second insulating layer **601** may cover a top surface of the molding layer **400**. The second insulating layer **601** may be an organic insulating layer. For example, the second insulating layer **601** may include an organic material such as a photo-imageable dielectric (PID) material. As another example, the second insulating layer **601** may include a solder resist material or an Ajinomoto build-up film. The second redistribution substrate **600** may include a plurality of second insulating layers **601**. The second insulating layers **601** may be stacked on the molding layer **400**. The second insulating layers **601** may include the same material. The second insulating layers **601** may include different materials. An interface between the second insulating layers **601** adjacent to each other may not be distinguished, but is not limited thereto. For example, an interface between the second insulating layers **601** adjacent to each other may be distinguished. The number of the second insulating layers **601** may be variously modified. The second insulating layers **601** may be transparent.
(43) The second redistribution patterns **630** may be laterally spaced apart from one another and may be electrically separated from one another. Each of the second redistribution patterns **630** may include a second via part and a second wiring part. The second via part of each of the second redistribution patterns **630** may be provided in a corresponding second insulating layer **601**. The second wiring part of each of the second redistribution patterns **630** may be provided between the second insulating layers **601**. The second via part of each of the second redistribution patterns **630** may be connected to the second wiring part without an interface. The second redistribution patterns **630** may include a metal such as copper.
(44) The second redistribution patterns **620** may include second lower redistribution patterns **631** and second upper redistribution patterns **632**. The second lower redistribution patterns **631** may be provided in an edge region of the second redistribution substrate **600** in a plan view. The second lower redistribution patterns **631** may be disposed on the conductive structures **350** to connect to the conductive structures **350**.
(45) The second upper redistribution patterns **632** may be disposed on the second lower redistribution patterns **631** and may be connected to the second lower redistribution patterns **631**. The second upper redistribution patterns **632** may be connected to the conductive structures **350** through the second lower redistribution patterns **631**. As illustrated in FIGS. **1**A and **1**B, the second upper redistribution patterns **632** may be provided in an edge region of the second redistribution substrate **600** in a plan view. The edge region of the second redistribution substrate **600** may be provided between side surfaces of the second redistribution substrate **600** and a center region of the second redistribution substrate **600** in a plan view.
(46) The second redistribution patterns **630** may include signal redistribution patterns and voltage supply redistribution patterns. For example, the signal redistribution patterns may function as data signal transmission paths between the first redistribution substrate **100** and the second redistribution pads **650**. The voltage supply redistribution patterns may function as voltage supply paths between the first redistribution substrate **100** and the second redistribution pads **650**. The

voltage may be a power supply voltage or a ground voltage. That is, the voltage supply redistribution patterns may include a ground voltage supply redistribution pattern and a power supply voltage supply redistribution pattern. The voltage supply redistribution patterns may be insulated from the signal redistribution patterns.

(47) The second seed patterns **635** may be respectively disposed on bottom surfaces of the second redistribution patterns **630**. For example, each of the second seed patterns **635** may be provided on a bottom surface and a side surface of the second via of the corresponding second redistribution pattern **620**, and may extend onto a bottom surface of the second wiring. The second seed patterns **635** may include a material different from that of the conductive structures **350** and the second redistribution patterns **630**. For example, the second seed patterns **635** may include a conductive seed material. The second seed patterns **635** may function as barrier layers to prevent diffusion of a material included in the second redistribution patterns **630**.

(48) The second redistribution pads **650** may be disposed on the second upper redistribution patterns **632** to respectively connect to the second upper redistribution patterns **632**. The second redistribution pads **650** may be laterally spaced apart from one another. As illustrated in FIGS. **1**A and **1**B, the second redistribution pads **650** may be provided in an edge region of the second redistribution substrate **600** in a plan view. As illustrated in FIG. **1**D, the second redistribution pads **650** may be electrically connected to the conductive structures **350** through the second redistribution patterns **630**. Thus, the second redistribution patterns **630** may be provided to electrically connect second redistribution pads **650** to the conductive structures **350** even when the second redistribution pads **650** are not vertically aligned with the conductive structure **350**. Accordingly, the arrangement of the second redistribution pads **650** may be designed more freely.

(49) A lower part of each of the second redistribution pads **650** may be provided in the uppermost second insulating layer **601**. An upper part of each of the second redistribution pads **650** may extend onto a top surface of the uppermost second insulating layer **601**. The upper part of each of the second redistribution pads **650** may have a greater width than the lower part of each of the second redistribution pads **650**. A top surface of each of the second redistribution pads **650** may be exposed through the uppermost second insulating layer **601**. The second redistribution pads **650** may include, for example, a metal such as copper.

(50) The number of the second redistribution patterns **630** that are stacked may be variously modified. For example, the second lower redistribution patterns **631** may be omitted, and the second upper redistribution patterns **632** may be disposed on the conductive structures **350**. As another example, second intermediate redistribution patterns may be further provided between the second lower redistribution patterns **631** and the second upper redistribution patterns **632**.

(51) An outer wall of the second redistribution substrate **600** may be aligned with an outer wall of the molding layer **400** and an outer wall of the first redistribution substrate **100**. The second redistribution substrate **600** may be electrically connected to the conductive structures **350**. An electrical connection to the second redistribution substrate **600** may include an electrical to at least one of the second redistribution patterns **630**.

(52) The first dummy conductive pattern **641**, the second dummy conductive pattern **642**, and the third dummy conductive pattern **643** may be provided between the second insulating layers **601**. One second insulating layer **601** may cover a top surface of the first dummy conductive pattern **641**, a top surface of the second dummy conductive pattern **642**, and a top surface of the third dummy conductive pattern **643**. As illustrated in FIGS. **1**A and **1**B, the first to third dummy conductive patterns **641**, **642**, and **643** may be provided in the center region of the second redistribution substrate **600** in a plan view. The first to third dummy conductive patterns **641**, **642**, and **643** may be spaced apart from the second redistribution patterns **630**. For example, the first to third dummy conductive patterns **641**, **642**, and **643** may be horizontally spaced apart from the second upper redistribution patterns **632**. The first to third dummy conductive patterns **641**, **642**, and **643** may be insulated from the second redistribution patterns **630** and from each other. A

thickness of each of the first to third dummy conductive patterns **641**, **642**, and **643** may be substantially the same as a thickness of the second upper redistribution patterns **632**. The same thicknesses, levels, and intervals of certain components may error ranges that may occur during a process. As another example, at least one of the first dummy conductive pattern **641**, the second dummy conductive pattern **642**, and the third dummy conductive pattern **643** may be electrically connected to the second redistribution patterns **630**.

(53) The first dummy conductive pattern **641** may be provided between the second dummy conductive pattern **642** and the third dummy conductive pattern **643**. As shown in FIG. **1**B, the first dummy conductive pattern **641** may have inner walls facing the second dummy conductive pattern **642** and outer walls facing the third dummy conductive pattern **643**. For example, inner walls of the second dummy conductive pattern **642** may have a rectangular shape in a plan view. Outer walls of the second dummy conductive pattern **642** may have a rectangular shape in a plan view. The shapes of the inner and outer walls of the second dummy conductive pattern **642** are not limited to the drawings shown and may be variously modified. For example, each of the outer and inner walls of the second dummy conductive pattern **642** may have a circular or polygonal shape.

(54) The second dummy conductive pattern **642** may be horizontally spaced apart from the first dummy conductive pattern **641**. For example, the second dummy conductive pattern **642** may be horizontally spaced apart from the outer walls of the first dummy conductive pattern **641**. The first dummy conductive pattern **641** and the second dummy conductive pattern **642** may be spaced apart from each other by a first interval A**1**. The first interval A**1** may be, for example, 1 μm to 5 mm. The second dummy conductive pattern **642** may be surrounded by the first dummy conductive pattern **641** in a plan view as shown in FIG. **1**B. First holes **641**H may be formed through the second dummy conductive pattern **642**. The first holes **641**H may pass through top and bottom surfaces of the second dummy conductive pattern **642**. The first holes **641**H may function as passages through which impurities are discharged in a base state in a process of forming the second dummy conductive pattern **642**. The base state is defined as a state in which by-products or impurities generated while forming the second dummy conductive pattern **642** remain. The shape and planar arrangement of the first holes **641**H may be variously modified. The second insulating layer **601** may pass through and fill the first holes **641**H, and in this regard the adjacent second insulating layers **601** may directly contact each other through the first holes **641**H.

(55) The third dummy conductive pattern **643** may be interposed between the first dummy conductive pattern **641** and the second upper redistribution patterns **632**. The third dummy conductive pattern **643** may be horizontally spaced apart from the second dummy conductive pattern **642** and the second upper redistribution patterns **632**. For example, the third dummy conductive pattern **643** may be horizontally spaced apart from the outer walls of the second dummy conductive pattern **642**. A second interval A**2** between the second dummy conductive pattern **642** and the third dummy conductive pattern **643** may be, for example, 1 μm to 5 mm. The third dummy conductive pattern **643** may surround the first dummy conductive pattern **641** in a plan view as shown in FIG. **1**B. Second holes **643**H may be formed through the third dummy conductive pattern **643**. The second holes **643**H may pass through top and bottom surfaces of the third dummy conductive pattern **643**. The second holes **643**H may function as passages through which impurities are discharged in a base state in a process of forming the third dummy conductive pattern **643**. The base state is defined as a state in which by-products or impurities generated while forming the third dummy conductive pattern **643** remain. The shape and planar arrangement of the second holes **643**H may be variously modified. The second insulating layer **601** may pass through and fill the second holes **643**H, and in this regard the adjacent second insulating layers **601** may directly contact each other through the second holes **643**H.

(56) The first to third dummy conductive patterns **641**, **642**, and **643** may include a metal such as copper. As another example, the first to third dummy conductive patterns **641**, **642**, and **643** may include tungsten or aluminum. Thermal expansion coefficients of the first to third dummy

conductive patterns **641**, **642**, and **643** may be different from that of the second insulating layers **601**. For example, the thermal expansion coefficients of the first to third dummy conductive patterns **641**, **642**, and **643** may be greater than that of the second insulating layers **601**.
(57) The marking metal layer **660** may be provided on the second dummy conductive pattern **642**. For example, the marking metal layer **660** may be provided on the top surface of the uppermost second insulating layer **601**. As shown in FIG. **1**C, the marking metal layer **660** may be provided on the center region of the second redistribution substrate **600**. The marking metal layer **660** may be horizontally spaced apart from the second redistribution pads **650**. The marking metal layer **660** may be electrically insulated from the second redistribution pads **650**, the second redistribution patterns **630**, and the first to third dummy conductive patterns **641**, **642**, and **643**. The marking metal layer **660** may include a marking part MK. The marking part MK may be provided on a top surface of the marking metal layer **660**. The marking part MK may indicate information about the semiconductor package **10**. The top surface of the marking metal layer **660** may not be covered by the uppermost second insulating layer **601**, and thus the marking part MK may be exposed.
(58) As another example, the marking metal layer **660** may be electrically connected to at least one of the second redistribution pads **650**, the second redistribution patterns **630**, and the first to third dummy conductive patterns **641**, **642**, and **643**.
(59) A thermal expansion coefficient of the marking metal layer **660** may be different from a thermal expansion coefficient of the second insulating layers **601**. The thermal expansion coefficient of the marking metal layer **660** may be greater than that of the second insulating layers **601**. As shown in FIG. **1**F, when the second redistribution substrate **600** does not include the first to third dummy conductive patterns **641**, **642**, and **643**, stress due to a difference in the thermal expansion coefficients between the second insulating layers **601** and the marking metal layer **660** may be concentrated on the edge region of the marking metal layer **660**. Cracks CR may be formed in the second redistribution substrate **600** by the stress. The crack CR may be formed in the uppermost second insulating layer **601**, and may overlap the edge region of the marking metal layer **660** or may be adjacent to the edge region of the marking metal layer **660**. The second redistribution substrate **600** may be damaged by the crack CR.
(60) According to example embodiments, as shown in FIGS. **1**A, **1**D, and **1**E, the first dummy conductive pattern **641** may be provided on a bottom surface of the edge region of the marking metal layer **660**. For example, sidewalls of the marking metal layer **660** may vertically overlap the first dummy conductive pattern **641**. The first dummy conductive pattern **641** may include a first part and a second part. The first part of the first dummy conductive pattern **641** may overlap the marking metal layer **660** in a plan view. The second part of the first dummy conductive pattern **641** may be spaced apart from the marking metal layer in a plan view. The second part of the first dummy conductive pattern **641** may be connected to the first part without an interface.
(61) The first dummy conductive pattern **641** may have a larger thermal expansion coefficient than that of the second insulating layers **601**. The difference in the thermal expansion coefficients between the marking metal layer **660** and the second insulating layers **601** may be offset or dispersed by the difference in the thermal expansion coefficients between the first dummy conductive pattern **641** and the second insulating layers **601**. Accordingly, a phenomenon in which stress due to the difference in the thermal expansion coefficients between the second insulating layers **601** and the marking metal layer **660** is concentrated on the marking metal layer **660** may be prevented.
(62) When one of the second dummy conductive pattern **642** and the third dummy conductive pattern **643** is omitted, a stress the difference in the thermal expansion coefficient between the first dummy conductive pattern **641** and the second insulating layers **601** may be applied to the first dummy conductive pattern **641**. In some example embodiments, the second dummy conductive pattern **642** and the third dummy conductive pattern **643** may be provided on both sides of the first dummy conductive pattern **641**, respectively. Each of the thermal expansion coefficient of the

second dummy conductive pattern **642** and the thermal expansion coefficient of the third dummy conductive pattern **643** may be greater than the thermal expansion coefficient of the second insulating layers **601**. Stress due to the difference in the thermal expansion coefficients between the first dummy conductive pattern **641** and the second insulating layers **601** on inner walls of the first dummy conductive pattern **641** may be offset or dispersed by the stress due to the difference in the thermal expansion coefficients between the second dummy conductive pattern **642** and the second insulating layers **601**. Accordingly, a phenomenon in which stress is concentrated on the inner walls of the first dummy conductive pattern **641** may be prevented. When the first interval A**1** is less than 1 μm, the stress may be applied to the inner walls of the first dummy conductive pattern **641**. In some example embodiments, when the first interval A**1** is 1 μm or more, the phenomenon in which the stress is concentrated on the inner walls of the first dummy conductive pattern **641** may be further prevented.
(63) Stress due to a difference in thermal expansion coefficients between the first dummy conductive pattern **641** and the second insulating layers **601** on the outer walls of the first dummy conductive pattern **641** may be offset or dispersed by stress due to a difference in thermal expansion coefficients between the third dummy conductive pattern **643** and the second insulating layers **601**. Accordingly, a phenomenon in which stress is concentrated on the outer walls of the first dummy conductive pattern **641** may be prevented. In some example embodiments, when the second interval A**2** is 1 μm or more, the phenomenon in which the stress is concentrated on the outer walls of the first dummy conductive pattern **641** may be further prevented.
(64) According to example embodiments, although an operation of the semiconductor package **10** is repeated, a phenomenon in which a crack (CR of FIG. **1**F) is formed in the second redistribution pattern **630** may be prevented. The operation is defined as a series of operation processes in a general device into which the semiconductor package **10** is inserted. Accordingly, durability and reliability of the semiconductor package **10** may be improved.
(65) When the first interval A**1** is greater than 500 μm or the second interval A**2** is greater than 500 μm, a size of the semiconductor package **10** may be excessively increased. According to example embodiments, the first interval A**1** may be between 1 μm and 500 μm, and the second interval A**2** may be between 1 μm and 500 μm. Accordingly, the semiconductor package **10** may be miniaturized and the phenomenon in which stress is concentrated on the outer walls of the first dummy conductive pattern **641** may be prevented.
(66) Hereinafter, with reference to FIG. **1**E, the first to third dummy conductive patterns **641**, **642**, and **643**, the marking metal layer **660**, and the second redistribution pads **650** according to example embodiments will be described in more detail.
(67) As shown in FIG. **1**E, the second redistribution substrate **600** may further include dummy seed patterns **645**. Dummy seed patterns **645** may be respectively disposed on the bottom surfaces of the first to third dummy conductive patterns **641**, **642**, and **643**. The dummy seed patterns **645** may include a conductive seed material. The dummy seed patterns **645** may include a metal material different from that of the first to third dummy conductive patterns **641**, **642**, and **643**. For example, the dummy seed patterns **645** may include titanium or an alloy of titanium and copper. A thickness of the dummy seed patterns **645** may be smaller than the thicknesses of the first to third dummy conductive patterns **641**, **642**, and **643**. The dummy seed patterns **645** may include the same material as the second seed patterns **635** and may have substantially the same thickness as the second seed patterns **635**. The dummy seed patterns **645** may function as barrier layers to prevent diffusion of metal included in the first to third dummy conductive patterns **641**, **642**, and **643**.
(68) Each of the second redistribution pads **650** may include a second seed pad **655** and a conductive pad **651**. The second seed pad **655** may be interposed between the corresponding second upper redistribution pattern **632** and the second redistribution pad **650**. The second seed pad **655** may further extend between the corresponding second upper redistribution pattern **632** and the uppermost second insulating layer **601**. The second seed pad **655** may include a conductive seed

material. The second seed pad **655** may function as a barrier layer to prevent diffusion of metal included in the conductive pad **651**.

(69) The conductive pad **651** may be disposed on the second seed pad **655**. The conductive pad **651** may include a material different from that of the second seed pad **655**. For example, the conductive pad **651** may include copper, and the second seed pad **655** may include titanium or an alloy of titanium and copper. A thickness of the conductive pad **651** may be greater than the thickness of the second seed pad **655**.

(70) The marking metal layer **660** may include a marking seed pattern **665** and a metal pattern **661**. The marking seed pattern **665** may be disposed on the uppermost second insulating layer **601**. The marking seed pattern **665** may include a conductive seed material. For example, the marking seed pattern **665** may include titanium or an alloy of titanium and copper. The marking seed pattern **665** may include the same material as that of the second seed pad **655**. The marking seed pattern **665** may have substantially the same thickness as the second seed pad **655**. The marking seed pattern **665** may function as a barrier layer to prevent diffusion of metal included in the marking metal layer **660**.

(71) The metal pattern **661** may be provided on the marking seed pattern **665**. A thickness of the metal pattern **661** may be greater than the thickness of the marking seed pattern **665**. The metal pattern **661** may include the same material as the conductive pad **651**. The metal pattern **661** may include a material different from that of the marking seed pattern **665**. For example, the metal pattern **661** may include copper. Alternatively, the conductive pattern and the marking seed pattern **665** may include the same metal. As another example, the marking metal layer **660** may not include the marking seed pattern **665**.

(72) FIG. **1**G is a view for explaining a marking metal layer and a second redistribution pad according to example embodiments and corresponds to an enlarged view of region “III” of FIG. **1**D.

(73) Referring to FIG. **1**G, each of the second redistribution pads **650** may include a second seed pad **655**, a conductive pad **651**, and a pad passivation layer **653**. The second seed pad **655** and the conductive pad **651** may be substantially the same as described in the example of FIG. **1**E. The pad passivation layer **653** may be disposed on a top surface of the conductive pad **651**. The pad passivation layer **653** may include a metal different from that of the conductive pad **651**. For example, the pad passivation layer **653** may include nickel, gold (Au), or an alloy thereof. The pad passivation layer **653** may be a single layer or multiple layers. A thickness of the pad passivation layer **653** may be smaller than the thickness of the conductive pad **651**. The pad passivation layer **653** may prevent damage (e.g., oxidation) of the conductive pad **651**.

(74) The marking metal layer **660** may include a marking seed pattern **665**, a metal pattern **661**, and a marking passivation layer **663**. The marking passivation layer **663** may be provided on a top surface of the metal pattern **661**. The marking passivation layer **663** may include a metal different from the metal pattern **661**. The marking passivation layer **663** may include the same material as the pad passivation layer **653**. For example, the pad passivation layer **653** may include nickel, gold (Au), or an alloy thereof. A thickness of the marking passivation layer **663** may be smaller than the thickness of the metal pattern **661**. The marking passivation layer **663** may be substantially the same as the thickness of the pad passivation layer **653**. The marking passivation layer **663** may prevent damage (e.g., oxidation) of the metal pattern **661**. The marking passivation layer **663** may be a single layer or multiple layers. For example, the marking passivation layer **663** may include an adhesive layer and a bonding layer on the adhesive layer. The bonding layer may include a metal different from the adhesive layer. For example, the bonding layer may include gold (Au) and an adhesive layer (Ni). As another example, one of the bonding layer and the adhesive layer may be omitted.

(75) Hereinafter, the marking metal layer according to the example embodiments will be described in more detail.

(76) FIG. **2**A is a plan view illustrating a marking metal layer according to example embodiments. FIG. **2**B is a cross-section taken along line IV-V line of FIG. **2**A.
(77) Referring to FIGS. **2**A and **2**B, the marking metal layer **660** may have the marking part MK. The marking part MK may be provided on the top surface of the marking metal layer **660**. The marking part MK may be visually distinguishable from other portions of the marking metal layer **660**, and may indicate information about the semiconductor package (**10** in FIG. **1**D) by representing a logo, letters, numbers, and/or barcodes. The marking metal layer **660** may have a first top surface **660***a***1** and a second top surface **660***a***2**. The first top surface **660***a***1** of the marking metal layer **660** may have a different roughness from that of the second top surface **660***a***2**. For example, the first top surface **660***a***1** of the marking metal layer **660** may have a smaller surface roughness than that of the second top surface **660***a***2**. Forming the first top surface **660***a***1** of the marking metal layer **660** may include melting at least a part of the marking metal layer **660** by, for example, irradiating a laser. The second top surface **660***a***2** of the marking metal layer **660** may not be exposed to the laser. Accordingly, the first top surface **660***a***1** of the marking metal layer **660** may be flat, and may have a smaller surface roughness than that of the second top surface **660***a***2** of the marking metal layer **660**. The marking part MK may be marked by a difference in surface roughness of the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660**. For example, the marking part MK may correspond to the first top surface **660***a***1** of the marking metal layer **660**.
(78) The first top surface **660***a***1** of the marking metal layer **660** may be provided at the same or different level as the second top surface **660***a***2** of the marking metal layer **660**. In the present disclosure, the level may indicate a vertical level, and a level difference may be measured in a direction perpendicular to the bottom surface of the marking metal layer **660**.
(79) The marking metal layer **660** may include the marking seed pattern **665** and the metal pattern **661**. The first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660** may correspond to the top surface of the metal pattern **661**.
(80) FIG. **2**C is a view for illustrating a marking metal layer according to example embodiments, and corresponds to a cross-section taken along line IV-V of FIG. **2**A.
(81) Referring to FIGS. **2**A and **2**C, the marking metal layer **660** may include the marking seed pattern **665**, the metal pattern **661**, and the marking passivation layer **663**. The first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660** may correspond to the top surface of the marking passivation layer **663**. The marking part MK may be provided on the top surface of the marking passivation layer **663**. The marking part MK may be visible due to a difference in surface roughness of the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660**.
(82) FIG. **2**D is a view for illustrating a marking metal layer according to example embodiments, and corresponds to a cross-section taken along line IV-V of FIG. **2**A.
(83) Referring to FIGS. **2**A and **2**D, the marking metal layer **660** may have the first top surface **660***a***1** and the second top surface **660***a***2**. The first top surface **660***a***1** of the marking metal layer **660** may have a greater surface roughness than that of the second top surface **660***a***2**. The first top surface **660***a***1** of the marking metal layer **660** may be provided at the same or different level as the second top surface **660***a***2**. The marking part MK may be visible due to the difference in surface roughness of the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660**.
(84) The marking metal layer **660** may include the marking seed pattern **665**, the metal pattern **661**, and the marking passivation layer **663**. The first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660** may correspond to the top surface of the marking passivation layer **663**. The marking part MK may be provided on the top surface of the marking passivation layer **663**.
(85) According to some example embodiments, the marking metal layer **660** may not include the

marking passivation layer **663**. In this case, the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660** may be formed on the top surface of the metal pattern **661**.
(86) FIG. **2**E is a view for illustrating a marking metal layer according to example embodiments, and corresponds to a cross-section taken along line IV-V of FIG. **2**A.
(87) Referring to FIGS. **2**A and **2**E, the marking metal layer **660** may have the first top surface **660***a***1** and the second top surface **660***a***2**. The first top surface **660***a***1** of the marking metal layer **660** may be provided at a level different from that of the second top surface **660***a***2**. For example, the first top surface **660***a***1** of the marking metal layer **660** may be positioned at a lower level than the second top surface **660***a***2**. A groove **669** may be formed on the second top surface **660***a***2** of the marking metal layer **660** to form the first top surface **660***a***1** of the marking metal layer **660**. The first top surface **660***a***1** of the marking metal layer **660** may correspond to a bottom surface of the groove **669**. The marking part MK may be visible due to the level difference between the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660**.
(88) As another example, the first top surface **660***a***1** of the marking metal layer **660** may be positioned at a higher level than the second top surface **660***a***2**. In this case, the second top surface **660***a***2** of the marking metal layer **660** may correspond to the bottom surface of the groove **669**.
(89) The second top surface **660***a***2** of the marking metal layer **660** may correspond to the top surface of the marking passivation layer **663**. The first top surface **660***a***1** of the marking metal layer **660** may be provided in the marking passivation layer **663**. As another example, the groove **669** may further penetrate at least a portion of the metal pattern **661**, and the first top surface **660***a***1** of the marking metal layer **660** may expose the metal pattern **661**. As another example, the marking metal layer **660** may not include the marking passivation layer **663**, and the first top surface **660***a***1** and the second top surface **660***a***2** of the marking metal layer **660** may correspond to the top surface of the metal pattern **661**.
(90) Hereinafter, semiconductor packages according to example embodiments will be described.
(91) FIG. **3**A is a diagram for illustrating a semiconductor package according to example embodiments.
(92) Referring to FIG. **3**A, a semiconductor package **10**A may be a lower package. The semiconductor package **10**A may include the first redistribution substrate **100**, the solder balls **500**, the semiconductor chip **200**, the conductive structures **350**, the molding layer **400**, and the second redistribution substrate **600**.
(93) The second redistribution substrate **600** may include the second insulating layers **601**, the second redistribution patterns **630**, the second redistribution pads **650**, dummy seed patterns **645**, the first to third dummy conductive patterns **641**, **642**, and **643**, the marking metal layer **660**, and a passivation layer **602**. The passivation layer **602** may be provided on the uppermost second insulating layer **601**. The passivation layer **602** may further cover sidewalls of the second redistribution pad **650** and sidewalls of the marking metal layer **660**. The passivation layer **602** may not cover the top surfaces of the second redistribution pads **650** and the top surface of the marking metal layer **660**. For example, the passivation layer **602** may have first upper openings, and the first upper openings may expose the top surfaces of the second redistribution pads **650**, respectively. The passivation layer **602** may have second upper openings, and the second upper openings may expose the top surface of the marking metal layer **660**. A top surface of the passivation layer **602** may be provided at the same level as or higher than the top surface of the marking metal layer **660** and the top surfaces of the second redistribution pad **650**. The passivation layer **602** may include a material different from that of the second insulating layers **601**. For example, the passivation layer **602** may include an insulating polymer. The passivation layer **602** may be opaque, but is not limited thereto.
(94) FIG. **3**B is a diagram for illustrating a semiconductor package according to example embodiments.
(95) Referring to FIG. **3**B, a semiconductor package **10**B may be a lower package. The

semiconductor package **10**B may include a first redistribution substrate **100**′, the solder balls **500**, the semiconductor chip **200**, the conductive structures **350**, the molding layer **400**, and the second redistribution substrate **600**. However, the semiconductor package **10**A may not include the conductive bumps **250** and the underfill layer **410** described in the example of FIG. **1**D.
(96) A first redistribution substrate **100**′ may include the first insulating layers **101**, the first redistribution patterns **130**, the first seed patterns **135**, first seed pads **145**, and first redistribution pads **140**. The first redistribution substrate **100**′ may not include the under bump patterns **120** described in the example of FIG. **1**D. The first redistribution substrate **100**′ may directly contact the semiconductor chip **200** and the molding layer **400**. For example, the uppermost first insulating layer **101** may directly contact the bottom surface of the semiconductor chip **200** and a bottom surface of the molding layer **400**. The first seed patterns **135** may be respectively provided on top surfaces of the first redistribution patterns **130**. The first seed patterns **135** in the uppermost first insulating layer **101** may directly connect to the chip pads **230** or the conductive structures **350**. The first via part of each of the uppermost first redistribution patterns **130** may vertically overlap the chip pads **230** or the conductive structures **350**.
(97) The first redistribution pads **140** may be provided on the bottom surface of the lowermost first insulating layer **101**. The first seed pads **145** may be respectively provided on top surfaces of the first redistribution pads **140**. The first redistribution pads **140** may function as pads of the solder balls **500**. For example, the solder balls **500** may be provided on the bottom surfaces of the first redistribution pads **140**, respectively.
(98) The semiconductor package **10**B may be manufactured by a chip-first process, but is not limited thereto.
(99) FIG. **3**C is a diagram for illustrating a semiconductor package according to example embodiments.
(100) Referring to FIG. **3**C, a semiconductor package **10**C may be a lower package. The semiconductor package **10**C includes the first redistribution substrate **100**, the solder balls **500**, the semiconductor chip **200**, a connection substrate **300**, the molding layer **400**, and the second redistribution substrate **600**.
(101) The first redistribution substrate **100** may be substantially the same as the first redistribution substrate **100** described in the example of FIG. **1**D. In this case, the semiconductor package **10**C may further include the conductive bumps **250** and the underfill layer **410**. As another example, the semiconductor package **10**C may include the first redistribution substrate **100**′ described in the example of FIG. **3**B.
(102) The connection substrate **300** may be disposed on the redistribution substrate **100**. A substrate hole **390** may penetrate through the connection substrate **300**. For example, the connection substrate **300** may be manufactured by forming the substrate hole **390** penetrating top and bottom surfaces of the printed circuit board. In a plan view, the substrate hole **390** may overlap a center portion of the first redistribution substrate **100**. The semiconductor chip **200** may be disposed in the substrate hole **390** of the connection substrate **300**. The semiconductor chip **200** may be disposed to be spaced apart from an inner wall of the connection substrate **300**.
(103) The connection substrate **300** may include conductive structures **350**′ and a base layer **310**. According to some example embodiments, the base layer **310** may include stacked layers. The base layer **310** may include an insulating material. For example, the base layer **310** may include a carbon-based material, a ceramic, or a polymer. The substrate hole **390** may pass through the base layer **310**. The conductive structures **350**′ may be provided in the base layer **310**. Each of the conductive structures **350**′ may be a metal pillar. The connection substrate **300** may further include first pads **351** and second pads **352**. The first pads **351** may be disposed on bottom surfaces of the conductive structures **350**′. The second pads **352** may be disposed on top surfaces of the conductive structures **350**′. The second pads **352** may be electrically connected to the first pads **351** through conductive structures **350**′. The conductive structures **350**′, first pads **351**, and second pads **352**

may include, for example, a metallic material such as copper, aluminum, tungsten, titanium, tantalum, iron, and alloys thereof.

(104) The semiconductor package **10**C may further include connection bumps **252** and an underfill pattern **420**. The connection bumps **252** may be disposed between the first redistribution substrate **100** and the connection substrate **300**. The connection bumps **252** may be interposed between the first pads **351** and the corresponding redistribution pads **150** to be connected to the first pads **351** and the corresponding redistribution pads **150**. The conductive structures **350′** may be electrically connected to the first redistribution substrate **100** by connection bumps **252**. The connection bumps **252** may include at least one of a solder ball, a bump, and a pillar. The connection bumps **252** may include a solder material. The underfill pattern **420** may be provided in a gap between the first redistribution substrate **100** and the connection substrate **300** to seal the connection bumps **252**. The underfill pattern **420** may include an insulating polymer. The underfill pattern **420** may include a material different from that of the molding layer **400**, but is not limited thereto.

(105) The molding layer **400** may be provided on the semiconductor chip **200** and the connection substrate **300**. The molding layer **400** may extend between the semiconductor chip **200** and the connection substrate **300**. According to example embodiments, an adhesive insulating film may be attached to a top surface of the connection substrate **300**, the top surface of the semiconductor chip **200**, and sidewalls of the semiconductor chip **200**, thereby forming the molding layer **400**. For example, Ajinomoto Build Up Film (ABF) may be used as the adhesive insulating film. As another example, the molding layer **400** may include an insulating polymer such as an epoxy-based polymer. As another example, the underfill pattern **420** may be omitted, and the molding layer **400** may further extend on a bottom surface of the connection substrate **300**. As another example, the underfill layer **410** may be omitted, and the molding layer **400** may further extend on the bottom surface of the semiconductor chip **200**.

(106) The second redistribution substrate **600** may be disposed on the molding layer **400** and the connection substrate **300**. The second redistribution substrate **600** may be substantially the same as described in the example of FIGS. **1**A to **1**E, or the example of FIG. **3**A. However, the second lower redistribution patterns **631** may extend into the molding layer **400** to be connected to the second pads **352**.

(107) FIG. **4** is a view for illustrating a semiconductor package according to example embodiments.

(108) Referring to FIG. **4**, a semiconductor package **30** may include a lower package **10′**, an upper package **20**, and connection solders **750**. The lower package **10′** may be substantially the same as the semiconductor package **10** described in the example of FIGS. **1**A to **1**E. As another example, the lower package **10′** may be substantially the same as the semiconductor package **10**A of FIG. **3**A, the semiconductor package **10**B of FIG. **3**B, or the semiconductor package **10**C of FIG. **3**C.

(109) The upper package **20** may include an upper substrate **710**, an upper semiconductor chip **720**, and an upper molding layer **740**. The upper substrate **710** may be disposed on a top surface of the second redistribution substrate **600** and may be spaced apart from the top surface of the second redistribution substrate **600**. The upper substrate **710** may be a printed circuit board (PCB) or a redistribution layer. First metal pads **711** and second metal pads **712** may be respectively disposed on a bottom surface and a top surface of the upper substrate **710**. Metal wires **715** may be provided in the upper substrate **710** to connect to the first metal pads **711** and the second metal pads **712**.

(110) The upper semiconductor chip **720** may be mounted on the upper substrate **710**. The upper semiconductor chip **720** may be a semiconductor chip of a different type from that of the semiconductor chip **200**. For example, the upper semiconductor chip **720** may be a memory chip, and the semiconductor chip **200** may be a logic chip. As another example, the upper semiconductor chip **720** may include a plurality of upper chips. The upper chips may be horizontally spaced apart from one another. Alternatively, the upper chips may be vertically stacked on the upper substrate **710**.

(111) Bonding wires **705** may be provided on a top surface of the upper semiconductor chip **720** to be electrically connected to chip pads **730** and the second metal pads **712** of the upper semiconductor chip **720**. The bonding wires **705** may include metal. As another example, the bonding wires **705** may be omitted, and the upper semiconductor chip **720** may be mounted on the upper substrate **710** by a flip-chip method. For example, upper bumps may be provided between the upper substrate **710** and the upper semiconductor chip **720** to be electrically connected to the upper substrate **710** and the upper semiconductor chip **720**. In this case, the chip pads **730** of the upper semiconductor chip **720** may be disposed on a bottom surface of the upper semiconductor chip **720**.

(112) The upper molding layer **740** may be provided on the upper substrate **710** and may cover the upper semiconductor chip **720**. The upper molding layer **740** may further cover the bonding wires **705**. The upper molding layer **740** may include an insulating polymer such as an epoxy-based molding compound.

(113) The upper package **20** may further include a heat dissipation structure **780**. The heat dissipation structure **780** may be disposed on the top surface of the upper semiconductor chip **720** and a top surface of the upper molding layer **740**. The heat dissipation structure **780** may include a heat sink, heat slug, and/or a thermal interface material (TIM) layer. The heat dissipation structure **780** may include, for example, a metal. As another example, the heat dissipation structure **780** may be omitted.

(114) FIGS. **5**A to **5**K are views for illustrating a method of manufacturing a semiconductor package according to example embodiments, and correspond to cross-sections taken along line I-II of FIG. **1**A. Hereinafter, descriptions that overlaps with the above-mentioned descriptions will be omitted. The description of FIGS. **5**A to **5**K will be described with reference to FIG. **1**A.

(115) Referring to FIG. **5**A, the under bump patterns **120**, the first insulating layer **101**, the first seed patterns **135**, and the first redistribution patterns **130** may be formed on a temporary substrate **900**.

(116) According to example embodiments, the under bump patterns **120** may be formed on the temporary substrate **900** by an electroplating process. The first insulating layer **101** may be formed on the temporary substrate **900**, and may cover sidewalls and top surfaces of the under bump patterns **120**. Openings **109** may be formed in the first insulating layer **101** to expose the under bump patterns **120**.

(117) Forming the first seed patterns **135** and the first redistribution patterns **130** may include forming a first seed layer in the openings **109** and on the top surface of the first insulating layer **101**, forming a resist layer on the first seed layer, performing an electroplating process using the first seed layer as an electrode, removing the resist layer to expose a part of the first seed layer, and etching the exposed part of the exposed first seed layer.

(118) The first redistribution patterns **130** may be formed in the openings **109** and under the resist layer by the electroplating process. Each of the first redistribution patterns **130** may include the first via part and the first wiring part. The first via part may be formed in the corresponding opening **109**. The first wiring part may be formed on the first via part and on the first insulating layer **101**. The first seed patterns **135** may be respectively formed on the bottom surfaces of the first redistribution patterns **130** by etching the first seed layer.

(119) Referring to FIG. **5**B, forming the first insulating layer **101**, forming the first seed patterns **135**, and forming the first redistribution patterns **130** may be repeatedly performed. Accordingly, stacked first insulating layers **101** and stacked first redistribution patterns **130** may be formed. The first seed patterns **135** may be respectively formed on the bottom surfaces of the first redistribution patterns **130**.

(120) The first seed pads **155** may be respectively formed in the openings **109** of the uppermost first insulating layer **101**. The first seed pads **155** may further extend on the top surface of the uppermost first insulating layer **101**. An electroplating process using the first seed pads **155** as

electrodes may be performed to form the first redistribution pads **150**. The first redistribution pads **150** may fill the openings **109** of the uppermost first insulating layer **101**. The first redistribution pads **150** may be connected to the corresponding first redistribution patterns **130**. Accordingly, the first redistribution substrate **100** may be manufactured. The first redistribution substrate **100** may include the first insulating layers **101**, the under bump patterns **120**, the first seed patterns **135**, the first redistribution patterns **130**, the first seed pads. **155**, and the first redistribution pads **150**.

(121) Referring to FIG. 5C, the conductive structures **350** may be formed on the top surface of the edge region of the first redistribution substrate **100**. For example, the conductive structures **350** may be provided on the corresponding first redistribution pads **150** to be connected to the first redistribution pads **150**. The conductive structures **350** may be formed by an electroplating process, but is not limited thereto.

(122) The semiconductor chip **200** may be mounted on the top surface of the center region of the first redistribution substrate **100**. Mounting the semiconductor chip **200** may include forming the conductive bumps **250** between the first redistribution substrate **100** and the semiconductor chip **200**. The conductive bumps **250** may connect to the corresponding first redistribution pads **150** and the chip pads **230**. Accordingly, the semiconductor chip **200** may be connected to the first redistribution substrate **100** through the conductive bumps **250**. The underfill layer **410** may be further formed between the first redistribution substrate **100** and the semiconductor chip **200**.

(123) The molding layer **400** may be formed on the top surface of the first redistribution substrate **100**, and may cover the semiconductor chip **200**. The molding layer **400** may cover sidewalls of the conductive structures **350** and expose top surfaces of the conductive structures **350**. For example, the top surface of the molding layer **400** may be provided at substantially the same level as the top surfaces of the conductive structures **350**.

(124) Referring to FIG. 5D, the second insulating layer **601** may be formed on the top surface of the molding layer **400**. The second insulating layer **601** may include a second lower insulating layer **601**L. The second lower insulating layer **601**L may expose top surfaces of the conductive structures **350**.

(125) The second seed patterns **635** may be formed on the top surface of the conductive structures **350**, and may extend on a top surface of the second lower insulating layer **601**L. The second seed patterns **635** may include second lower seed patterns **6351**.

(126) The second redistribution patterns **630** may be formed to cover the second lower seed patterns **6351**. The second redistribution patterns **630** may be the second lower redistribution patterns **631**. Forming the second lower seed patterns **6351** and the second lower redistribution patterns **631** may be the same as or similar to that described in the example of forming the first seed patterns **135** and the first redistribution patterns **130** of FIG. 5A, respectively.

(127) Thereafter, forming the second insulating layer **601** may be repeatedly performed to form a second intermediate insulating layer **601**M. The second intermediate insulating layer **601**M may cover the second lower insulating layer **601**L and the second lower redistribution patterns **631**. The second intermediate insulating layer **601**M may be formed by a coating process using a photosensitive insulating material. The second intermediate insulating layer **601**M may be patterned by an exposure and development process to form lower openings **608**. The lower openings **608** may penetrate the second intermediate insulating layer **601**M and expose the second lower redistribution patterns **631**.

(128) A first preliminary seed layer **645**P may be formed in the lower openings **608** and on a top surface of the second intermediate insulating layer **601**M. The first preliminary seed layer **645**P may include a conductive seed material.

(129) Referring to FIG. 5E, a first resist pattern **810** may be formed on the first preliminary seed layer **645**P. The first resist pattern **810** may have first guide holes **819** and second guide holes **818**. The first guide holes **819** and the second guide holes **818** may expose the first preliminary seed layer **645**P. The first guide holes **819** may be connected to the lower openings **608**. The second

guide holes **818** may be spaced apart from the first guide holes **819** and the lower openings **608** in a plan view.
(130) The second upper redistribution patterns **632** may be formed in the lower openings **608** and the first guide holes **819**. Each of the second upper redistribution patterns **632** may include the second via part and the second wiring part connected to each other. The second via part of each of the second upper redistribution patterns **632** may be formed in a corresponding lower opening **608**. The second via part of each of the second upper redistribution patterns **632** may be formed on the second via part and may extend onto a top surface of the second intermediate insulating layer **601**M. The second redistribution patterns **630** may include the second lower redistribution patterns **631** and the second upper redistribution patterns **632**.
(131) The second upper redistribution patterns **632** may be disposed in the second guide holes **818**, and thus a separate planarization process may not be required. Accordingly, the manufacturing process of the semiconductor package may be simplified.
(132) The first dummy conductive pattern **641**, the second dummy conductive pattern **642**, and the third dummy conductive pattern **643** may be formed in the second guide holes **818**. The first to third dummy conductive patterns **641**, **642**, and **643** may be laterally spaced apart from one another. The first to third dummy conductive patterns **641**, **642**, and **643** may be spaced apart from the second upper redistribution patterns **632**.
(133) The first to third dummy conductive patterns **641**, **642**, and **643** may be disposed in the corresponding second guide holes **818**, and thus a separate planarization process may not be required. Accordingly, the manufacturing process of the semiconductor package may be simplified.
(134) The first to third dummy conductive patterns **641**, **642**, and **643** may be formed by a single process with the second upper redistribution patterns **632**. Forming the second upper redistribution patterns **632** and the first to third dummy conductive patterns **641**, **642**, and **643** may include performing an electroplating process using the first preliminary seed layer **645**P as an electrode.
(135) Referring to FIG. **5**F, the first resist pattern **810** may be removed to expose a first part of the first preliminary seed layer **645**P. The first resist pattern **810** may be removed by a strip process. Second parts of the first preliminary seed layer **645**P may be provided on bottom surfaces of the second upper redistribution patterns **632**. Third parts of the first preliminary seed layer **645**P may be provided on the bottom surfaces of the first to third dummy conductive patterns **641**, **642**, and **643**.
(136) Referring to FIG. **5**G, the exposed first part of the first preliminary seed layer **645**P may be removed to form second upper seed patterns **6352** and dummy seed patterns **645**. Removing the first part of the first preliminary seed layer **645**P may be performed by an etching process. The second parts and third parts of the first preliminary seed layer **645**P may not be exposed to the etching process. After the etching process, the second parts of the remaining first preliminary seed layer **645**P may form the second upper seed patterns **6352** that are separated from one another. The second upper seed patterns **6352** may be respectively provided on the bottom surfaces of the second upper redistribution patterns **632**. After the etching process, the remaining third parts of the first preliminary seed layer **645**P may form the dummy seed patterns **645** separated from one another. The dummy seed patterns **645** may be provided on the bottom surfaces of the first to third dummy conductive patterns **641**, **642**, and **643**, respectively. The second seed patterns **635** may include the second lower seed patterns **6351** and second upper seed patterns **6352**.
(137) Referring to FIG. **5**H, a second upper insulating layer **601**U may be formed on the second intermediate insulating layer **601**M, the second upper redistribution patterns **632**, and the first to third dummy conductive patterns **641**, **642**, and **643**. The second upper insulating layer **601**U may include the same material as the second intermediate insulating layer **601**M. The second upper insulating layer **601**U may be patterned by an exposure and development process to form upper openings **609**. The upper openings **609** may expose the second upper redistribution patterns **632**. The upper openings **609** may not expose the first to third dummy conductive patterns **641**, **642**, and

643.
(138) A second preliminary seed layer **655**P may be formed in the upper openings **609** and on the second upper insulating layer **601**U. The second preliminary seed layer **655**P may cover the exposed second redistribution pads **650**. The second preliminary seed layer **655**P may be spaced apart from the first to third dummy conductive patterns **641**, **642**, and **643**. The second preliminary seed layer **655**P may include a conductive seed material.
(139) A second resist pattern **820** may be formed on the second preliminary seed layer **655**P. The second resist pattern **820** may have third guide holes **829** and fourth guide holes **828**. The third guide holes **829** and the fourth guide hole **828** may expose the second preliminary seed layer **655**P. The third guide holes **829** may be connected to the upper openings **609**. The fourth guide hole **828** may be spaced apart from the lower openings **608** and the third guide holes **829** in a plan view.
(140) The second redistribution pads **650** may be formed in the upper openings **609** and corresponding third guide holes **829**. The lower part of each of the second redistribution pads **650** may be formed in a corresponding upper opening **609**. The upper part of each of the second redistribution pads **650** may be formed under a corresponding third guide hole **829**.
(141) The metal pattern **661** may be formed in the third guide holes **829**. The metal pattern **661** may not be provided in the upper opening **609**. The metal pattern **661** may be formed with the second redistribution pads **650**. by a single process Forming the second redistribution pads **650** and the metal pattern **661** may include performing an electroplating process using the second preliminary seed layer **655**P as an electrode. The metal pattern **661** may be spaced apart from the second redistribution pads **650**.
(142) The second redistribution pads **650** may be disposed in the third guide holes **829** and the metal pattern **661** may be disposed in the fourth guide hole **828**, and thus a separate planarization process may not be required. Accordingly, the manufacturing process of the semiconductor package may be simplified.
(143) Referring to FIG. **5**I, the second resist pattern **820** may be removed to expose a first part of the second preliminary seed layer **655**P. The removal of the second resist pattern **820** may be performed by a strip process. Second parts of the second preliminary seed layer **655**P may be provided on bottom surfaces of the second redistribution pads **650**. A third part of the second preliminary seed layer **655**P may be provided on bottom surface of the metal pattern **661**.
(144) Referring to FIG. **5**J, the exposed first part of the second preliminary seed layer **655**P may be removed to form the second seed pad **655** and the marking seed pattern **665**. The removal of the first part of the second preliminary seed layer **655**P may be performed by an etching process. The second parts and the third part of the second preliminary seed layer **655**P may not be exposed to the etching process. After the etching process, the second parts of the remaining second preliminary seed layer **655**P may form the plurality of second seed pads **655** separated from one another. The second seed pads **655** may be respectively provided on bottom surfaces of the conductive pads **651**. Accordingly, the second redistribution pads **650** may be formed. The second redistribution patterns **630** may include the second seed pads **655** and the conductive pads **651**.
(145) After the etching process, the third part of the remaining second preliminary seed layer **655**P may form the marking seed pattern **665**. The marking seed pattern **665** may be provided on the bottom surface of the metal pattern **661**. Accordingly, the marking metal layer **660** may be formed. The marking metal layer **660** may include the marking seed pattern **665** and the metal pattern **661**.
(146) Manufacturing of the second redistribution substrate **600** may be completed by the examples described so far. The second redistribution substrate **600** may include the second insulating layers **601**, the second redistribution patterns **630**, the second seed patterns **635**, and the first to third dummy conductive patterns **641**, **642**, and **643**, the dummy seed patterns **645**, second redistribution pads **650**, and the marking metal layer **660**. The second insulating layers **601** may include the stacked second lower insulating layer **601**L, the second intermediate insulating layer **601**M, and the second upper insulating layer **601**U.

(147) Referring to FIG. 5K, the temporary substrate **900** may be removed to expose the bottom surface of the first redistribution substrate **100**. For example, the bottom surface of the lowermost first insulating layer **101** and the bottom surfaces of the under bump patterns **120** may be exposed.
(148) Referring back to FIG. 1D, the solder balls **500** may be respectively formed on the bottom surfaces of the under bump patterns **120**. The manufacturing of the semiconductor package **10** may be completed by the examples described so far.
(149) According to example embodiments, the redistribution substrate may include the marking metal layer and the dummy conductive pattern. The dummy conductive pattern may vertically overlap the sidewalls of the marking metal layer. Accordingly, the damage to the redistribution substrate may be prevented. The semiconductor package may have improved reliability and durability.
(150) While aspects of example embodiments have been particularly shown and described, it will be understood variations in form and detail may be made therein without departing from the spirit and scope of the attached claims.

Claims

1. A semiconductor package comprising: a first redistribution substrate; a semiconductor chip provided on the first redistribution substrate; a molding layer provided on the first redistribution substrate and the semiconductor chip; and a second redistribution substrate provided on the molding layer, wherein the second redistribution substrate comprises: redistribution patterns spaced apart from one another; a first dummy conductive pattern spaced apart from the redistribution patterns; an insulating layer provided on the first dummy conductive pattern; and a marking metal layer provided on the insulating layer and spaced apart from the first dummy conductive pattern, and wherein sidewalls of the marking metal layer overlap the first dummy conductive pattern along a vertical direction perpendicular to an upper surface of the first redistribution substrate.
2. The semiconductor package of claim 1, wherein the second redistribution substrate further comprises: a second dummy conductive pattern spaced apart from an inner wall of the first dummy conductive pattern; and a third dummy conductive pattern spaced apart from an outer wall of the first dummy conductive pattern.
3. The semiconductor package of claim 2, wherein the marking metal layer is electrically insulated from the first dummy conductive pattern, the second dummy conductive pattern and the third dummy conductive pattern.
4. The semiconductor package of claim 2, wherein the marking metal layer overlaps the first dummy conductive pattern along the vertical direction.
5. The semiconductor package of claim 4, wherein the marking metal layer is spaced apart from the third dummy conductive pattern along the vertical direction.
6. The semiconductor package of claim 1, further comprising conductive structures provided between the first redistribution substrate and the second redistribution substrate, wherein the second redistribution substrate further comprises second redistribution patterns connected to the conductive structures, and wherein the first dummy conductive pattern and the marking metal layer are insulated from the second redistribution patterns.
7. The semiconductor package of claim 6, further comprising redistribution pads provided on the second redistribution patterns and electrically connected to the second redistribution patterns, wherein the redistribution pads are provided in an edge region of the second redistribution substrate along the vertical direction, and wherein the marking metal layer is provided in a center region of the second redistribution substrate along the vertical direction.
8. The semiconductor package of claim 7, wherein the redistribution pads are insulated from the marking metal layer.
9. The semiconductor package of claim 1, wherein a top surface of the marking metal layer is

exposed through the second redistribution substrate.
10. The semiconductor package of claim 9, wherein the top surface of the marking metal layer comprises: a first top surface having a first surface roughness; and a second top surface having a second surface roughness that is different from the first surface roughness.
11. A semiconductor package comprising: a first redistribution substrate; a semiconductor chip provided on the first redistribution substrate; a molding layer provided on the first redistribution substrate and the semiconductor chip; and a second redistribution substrate provided on the molding layer, wherein the second redistribution substrate comprises: a redistribution pattern; a first dummy conductive pattern insulated from the redistribution pattern; a second dummy conductive pattern insulated from the redistribution pattern; a third dummy conductive pattern insulated from the redistribution pattern; and a marking metal layer provided on the second dummy conductive pattern, wherein the first dummy conductive pattern is provided between the second dummy conductive pattern and the third dummy conductive pattern, and wherein the marking metal layer overlaps a first portion of the first dummy conductive pattern along a vertical direction perpendicular to an upper surface of the first redistribution substrate, and is offset from a second portion of the first dummy conductive pattern along the vertical direction.
12. The semiconductor package of claim 11, wherein the third dummy conductive pattern is provided between the redistribution pattern and the second dummy conductive pattern along a horizontal direction parallel to the upper surface of the first redistribution substrate.
13. The semiconductor package of claim 12, further comprising redistribution pads provided on the and electrically connected to the redistribution pattern, wherein the redistribution pads are spaced apart from the marking metal layer along the horizontal direction.
14. The semiconductor package of claim 11, wherein the marking metal layer is insulated from each of the first dummy conductive pattern, the second dummy conductive pattern, the third dummy conductive pattern, and the redistribution pattern.
15. The semiconductor package of claim 11, wherein a plurality of first holes are formed through the first dummy conductive pattern, and wherein a plurality of second holes are formed through the third dummy conductive pattern.
16. A semiconductor package comprising: a first redistribution substrate comprising a first insulating layer, a first seed pattern, and a first redistribution pattern; a solder ball provided on a bottom surface of the first redistribution substrate; a semiconductor chip provided on a top surface of the first redistribution substrate; conductive structures provided on the top surface of the first redistribution substrate and spaced apart from the semiconductor chip along a horizontal direction parallel to an upper surface of the first redistribution substrate; a molding layer provided between the semiconductor chip and the conductive structures, and on the semiconductor chip; and a second redistribution substrate provided on the molding layer, wherein the second redistribution substrate comprises: second redistribution patterns electrically connected to the conductive structures; second redistribution pads provided on and electrically connected to the second redistribution patterns; a dummy conductive pattern spaced apart from the second redistribution patterns along the horizontal direction; and a marking metal layer spaced apart from the second redistribution pads along the horizontal direction, wherein the dummy conductive pattern comprises a first dummy conductive pattern, a second dummy conductive pattern, and a third dummy conductive pattern that are spaced apart from one another along the horizontal direction, wherein the marking metal layer overlaps a first portion of the first dummy conductive pattern and the second dummy conductive pattern along a vertical direction perpendicular to the upper surface of the first redistribution substrate, and wherein the marking metal layer is offset from a second portion of the first dummy conductive pattern and the third dummy conductive pattern along the vertical direction.
17. The semiconductor package of claim 16, wherein the marking metal layer is electrically insulated from each of the first dummy conductive pattern, the second dummy conductive pattern and the third dummy conductive pattern.

18. The semiconductor package of claim 16, wherein the marking metal layer comprises a marking seed pattern having a metal pattern, wherein one of the second redistribution pads comprises: a seed pad; and a conductive pad provided on the seed pad, wherein the marking seed pattern and the seed pad comprise a first common material, and wherein the metal pattern and the conductive pad comprise a second common material.

19. The semiconductor package of claim 16, wherein the second redistribution pads are provided in an edge region of the second redistribution substrate in a plan view, and wherein the marking metal layer is provided in a center region of the second redistribution substrate in the plan view.

20. The semiconductor package of claim 16, wherein the first dummy conductive pattern is provided between the second dummy conductive pattern and the third dummy conductive pattern, wherein a first interval between the first dummy conductive pattern and the second dummy conductive pattern is 1 μm to 5 mm, and wherein a second interval between the first dummy conductive pattern and the third dummy conductive pattern is 1 μm to 5 mm.